

Low Noise Anomaly

Team members:

M Taufiqul Huda <M Taufiqul Huda> [Team Leader]

Raditya Eka Putra

Elijah Teti Johnson

Dzaki Ardiansyah

Adam Ghieh

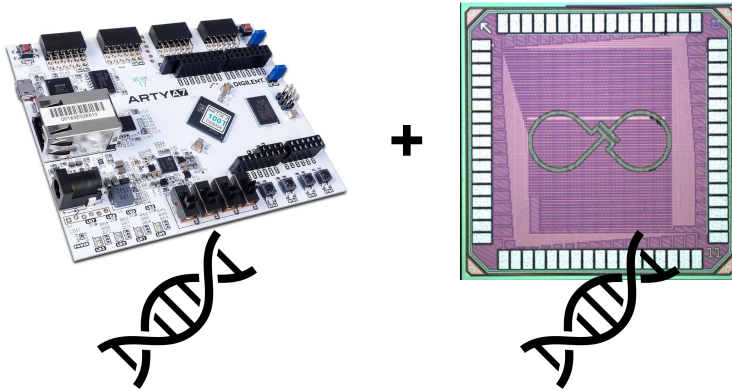
Team background

- Academic Experience
 - M Taufiqul Huda, Raditya Eka Putra & Dzaki Ardiansyah: Bachelor in Electrical Engineering, Institut Teknologi Sepuluh Nopember
 - Elijah Teti Johnson : Undergraduate Electrical Engineering at Columbia University
 - Adam Ghieh : Undergraduate Electrical Engineering at Concordia University
- Work Experience
 - M Taufiqul Huda : Analog IC Lab Intern at NTUST & Chipalooza Participant 2024
 - Raditya Eka Putra : Universalization of IC Design from CASS (IEEE) Participant 2024

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Project Background and Goal

- The MOSbius architecture integrates analog and digital circuits for transistor-level learning and direct simulation from schematic designs.
- It utilizes a Field Programmable Mixed-Signal Array (FPMA) to emulate analog and digital blocks, offering flexibility for various circuit configurations.



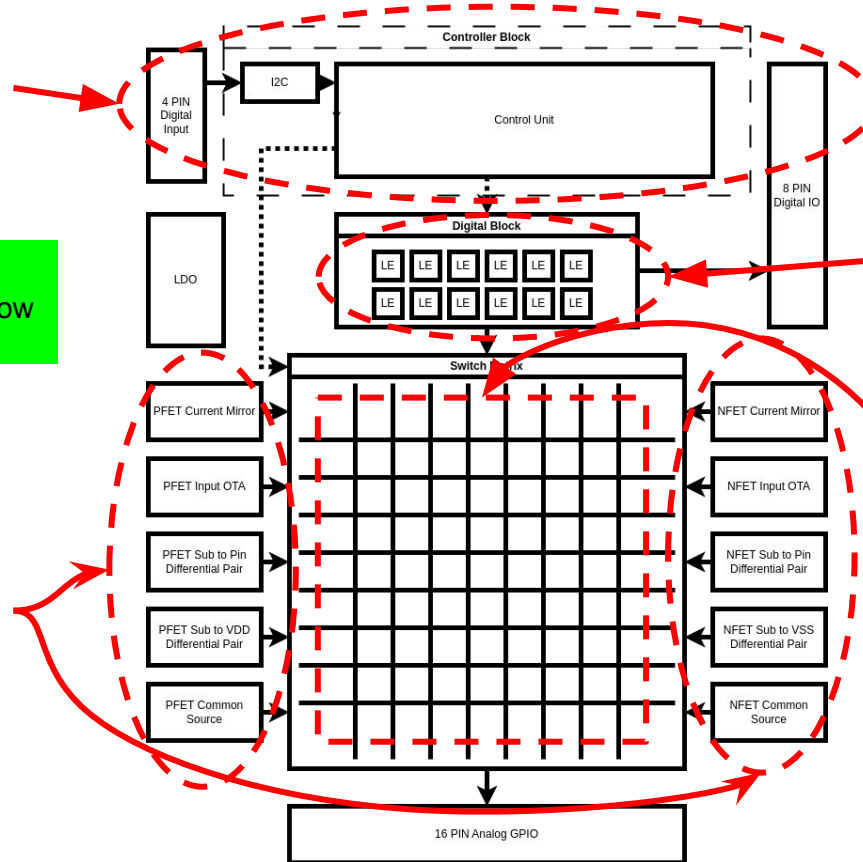
MIXED-SIGNAL MOSBIUS

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Controller Block : For control the switch matrix register and Logic Element register also programmable using I2C communication

Design Method :
Analog >>> Custom Analog Flow
Digital >>> Digital Flow

Analog Block : Contain lot of individual FET for flexibility of analog design also contain basic block of current mirror and OTA.

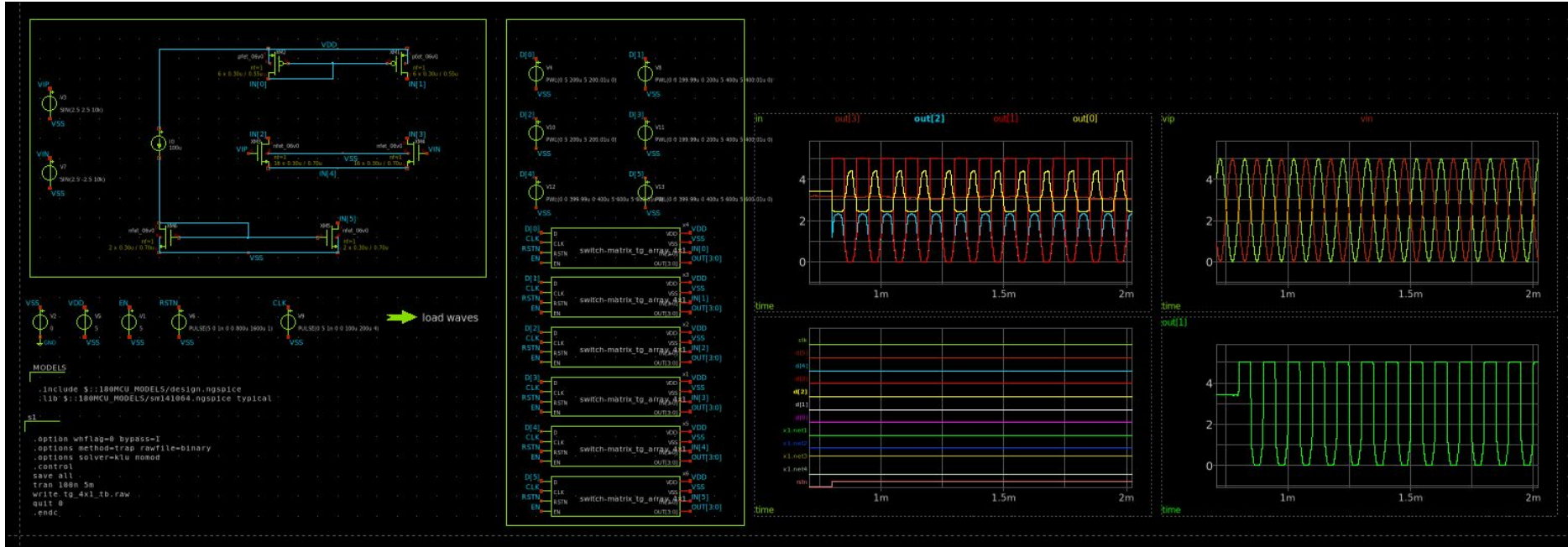


3.3 Volt Supply

Digital Block : Contain 16 digital Logic Element for simple digital logic for AMS design

AMS Switch Matrix : Bus and switch which connected to pin of matrix, work for digital also for analog

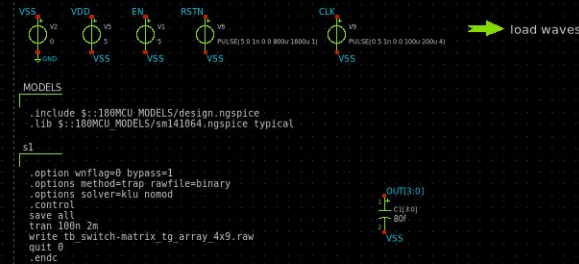
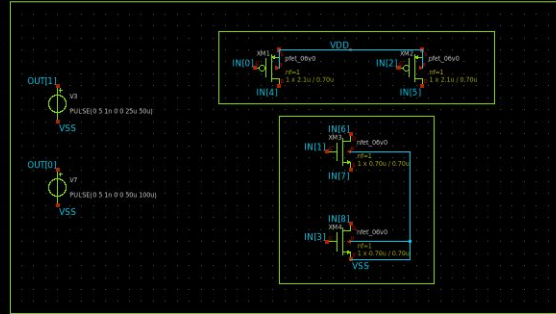
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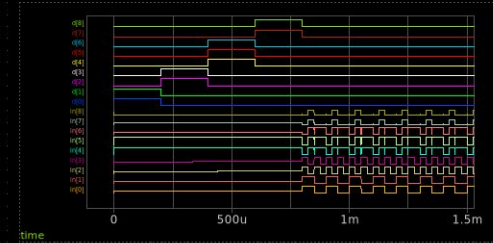
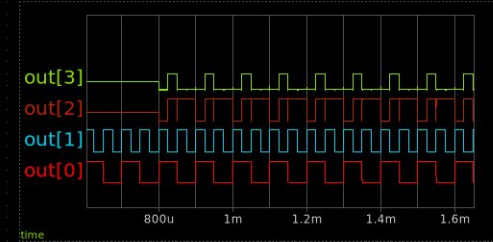
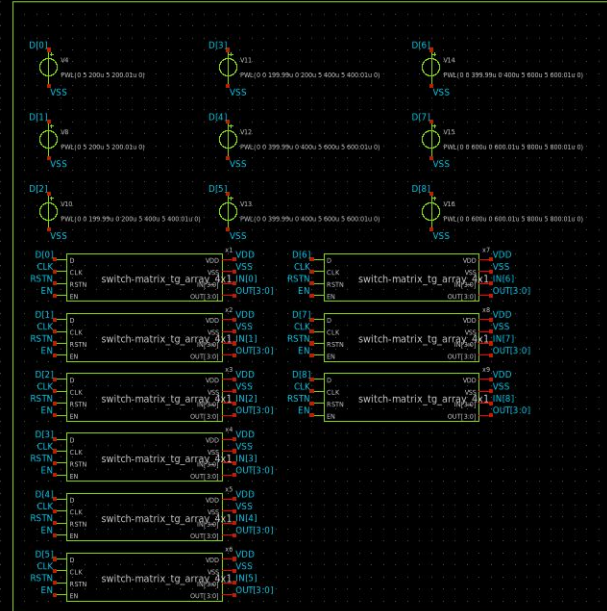
5T-OTA Mock Testbench Connected to Switch Matrix

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2-NAND



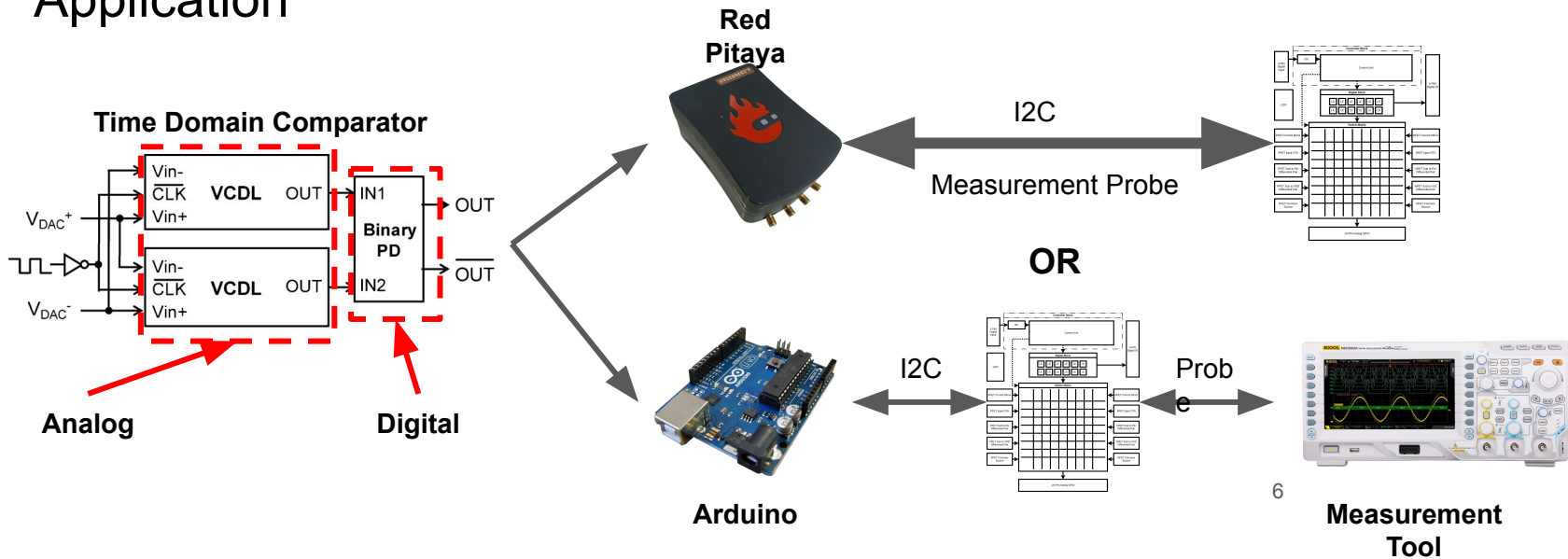
SWITCH MATRIX AND 4 BUS



2-NAND Mock Testbench Connected to Switch Matrix

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Application



Reference

[1] P. Chow, P. G. Gulak and P. Chow, "A Field-Programmable Mixed-Analog-Digital Array," *Third International ACM Symposium on Field-Programmable Gate Arrays*, Napa Valley, CA, USA, 1995, pp. 104-109, doi: 10.1109/FPGA.1995.242048.

Project Repository

<https://github.com/mthudaa/sscs-chipathon2025-LNA>